LESSON 1.2

Math is Exploring and Thinking

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can make sense of a problem, plan a strategy, and use fractions of a whole number to compare quantities.

WHY THIS MATTERS Doing math is exploring: you make sense of a problem, plan a strategy, check your progress, and try new paths when you hit a dead end. Today's problems — walking distances and skyscraper heights — reward thinkers, not just calculators. Try this one: a new building will be 3/4 as tall as a tower that is 400 meters tall. Plan a strategy to find how tall the new building will be.

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
quantity *Spanish: cantidad*	An amount in a problem — a number with a meaning attached, like 540 meters.	540 meters — the number 540 AND what it measures.
relationship *Spanish: relación*	How two quantities compare or connect to each other.	1 hour → 48 km, 2 hours → 96 km: as one grows, so does the other.
strategy *Spanish: estrategia*	A plan for how to solve a problem before you start computing.	"I will draw a tape diagram first, then divide" — chosen before computing.
reasonable *Spanish: razonable*	A solution that makes sense in the context of the problem.	5/6 of 540 must be under 540, so 450 is reasonable and 650 is not.
persevere *Spanish: perseverar*	To keep working on a problem, checking your progress and adjusting your plan when you get stuck.	Stuck on the tram problem → try a tape diagram instead → keep going.

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

A bowl is full of peas — far too many to count one at a time. About how many peas are in the bowl?

1. I do not count every pea — I look for a group I CAN count, like one small cluster of about 10.
2. Then I ask how many of those clusters would cover the whole bowl. I count about 12.
3. 12 clusters of about 10 is 12 × 10 = 120, so my estimate is about 120 peas — a reasoned answer, close enough to be useful.

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

105.76 = 100 + 5.76.

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- To know _____, I first need _____, because _____ depends on it.
- I know that _____ .
- I don't know _____ .
- When I'm stuck, I can _____ .

Word bank quantity | relationship | strategy | reasonable | persevere

WATCH OUT!

A common mistake

Expecting a fraction of a number to be bigger than the number

LESSON 1.2

Math is Exploring and Thinking

Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Apply it

real-world problems

1 **Circle the letter of the best answer. Show how you know.**

Which of these is a correct way to decompose 105.76?

(A) 10 + 5.76
(B) 100 + 5.76
(C) 100 + 5.076
(D) 105 + 7.6

HINT Add each option and compare to 105.76.

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

Which is another correct decomposition of 105.76?

(A) 106 − 0.34
(B) 110 − 4.24
(C) 110 − 5.24
(D) 110 + 4.24

HINT Start at 110. How far down is 105.76?

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

Deon walked a shorter distance than Miguel, and Evelyn walked a longer distance than Miguel. Who walked the shortest distance?

(A) Deon
(B) Miguel
(C) Evelyn
(D) It cannot be determined

HINT Put Miguel in the middle and place the other two around him.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

The London building will be 5/6 as tall as One World Trade Center, which is 540 meters. How tall will the London building be?

(A) 90 meters
(B) 450 meters
(C) 534 meters
(D) 648 meters

HINT First find one sixth of 540.

Rewrite each fraction _____

Work it out _____

Simplify _____

1. **Same size pieces** *Common denominator first.*
2. **Operate** *Only then add or subtract.*
3. **Simplify** *Divide out common factors.*

PART 2 Explain and correct

reasoning

5 **Answer in complete sentences.**

In the Comparing Walks problem, name one thing you KNOW and one thing you DON'T know.

SENTENCE STARTER I know that ___ .
I don't know ___ .

WRITE YOUR ANSWER

6 **Answer in complete sentences.**

What can you do if you're stuck on a problem?

SENTENCE STARTER When I'm stuck, I can ___ .
Another strategy I could try is ___ .

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

Pick one card you put under 'We DON'T know this yet.' What would you have to figure out first to move it to the known column, and which quantity does it depend on?

SENTENCE STARTER To know ___, I first need ___, because ___ depends on it.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 1.2

Math is Exploring and Thinking

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Circle the letter of the best answer. Show how you know.**

You calculate 5/6 × 540 and get 650 meters. What does the checking step of your plan tell you?

(A) The answer is fine because it is close to 540

(B) The answer is fine because 650 is a bigger number than 5/6

(C) The answer is unreasonable — 5/6 of 540 must be LESS than 540

(D) Checking is only needed when the problem asks for it

SHOW YOUR WORK

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

The Rio de Janeiro building will be 9/10 as tall as One World Trade Center (540 meters). How tall will it be?

(A) 54 meters

(B) 486 meters

(C) 594 meters

(D) 4,860 meters

Rewrite each fraction _____

Work it out _____

Simplify _____

3 **Circle the letter of the best answer. Show how you know.**

The Tokyo building will be 6/5 as tall as One World Trade Center (540 meters). How tall will it be?

(A) 108 meters
(B) 432 meters
(C) 450 meters
(D) 648 meters

Rewrite each fraction _____

Work it out _____

Simplify _____

4 **Circle the letter of the best answer. Show how you know.**

Without calculating, which new building must be TALLER than One World Trade Center, and why?

(A) London, because 5/6 has a 6 in it
(B) Tokyo, because 6/5 is greater than 1
(C) Rio de Janeiro, because 9/10 is close to 1
(D) None — a fraction of 540 is always less than 540

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

5 **Answer in complete sentences.**

Miguel walked 2 miles. Deon walked less than Miguel, and Evelyn walked more than Miguel. Choose possible distances for Deon and Evelyn and explain how you checked both relationships.

WRITE YOUR ANSWER

6 **Answer in complete sentences.**

The lesson says we check on our progress toward a solution and adjust our plan as needed. Describe one way YOU can check on your progress while solving a problem.

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

Pick one card you put under 'We DON'T know this yet.' What would you have to figure out first to move it to the known column, and which quantity does it depend on?

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LESSON 1.2

Math is Exploring and Thinking

Challenge · Extension

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Write the letter of the matching item on each line.**

Match each fraction of 540 to a shortcut that computes it.

_____ 9/10 of 540

_____ 6/5 of 540

_____ 5/6 of 540

_____ 1/2 of 540

(A) 540 − 54

(B) 540 + 108

(C) 540 − 90

(D) 540 ÷ 2

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

Which set shows FIVE different correct decompositions of 105.76?

(A) 100 + 5.76, 105 + 0.76, 110 − 4.24, 52.88 × 2, 211.52 ÷ 2

(B) 100 + 5.76, 105 + 7.6, 110 − 4.24, 52.88 × 2, 211.52 ÷ 2

(C) 100 + 5.76, 105 + 0.76, 110 + 4.24, 52.88 × 2, 211.52 ÷ 2

(D) 100 + 5.76, 105 + 0.76, 110 − 4.24, 52.88 × 3, 211.52 ÷ 2

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

Without computing the exact heights, order the three new buildings from shortest to tallest. (London: 5/6 of 540 m, Rio de Janeiro: 9/10 of 540 m, Tokyo: 6/5 of 540 m.)

(A) Rio de Janeiro, London, Tokyo

(B) Tokyo, Rio de Janeiro, London

(C) London, Tokyo, Rio de Janeiro

(D) London, Rio de Janeiro, Tokyo

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

4 **Answer in complete sentences.**

A classmate says, "In math, only the final answer matters." Using this lesson, write what you would say back.

WRITE YOUR ANSWER

5 **Answer in complete sentences.**

Write your own Comparing Walks problem: describe three people using only relationships (shorter than, longer than), so a reader can put them in order without any exact distances. Then give one set of values that fits.

WRITE YOUR ANSWER

6 **Answer in complete sentences.**

Apply — City Blocks. A city block has 12 buildings. The block to the north is planned to hold 2/3 as many buildings, and the block to the east is planned to hold 3/2 as many. WITHOUT computing, say which block will hold the most buildings and how you know. Then find the number of buildings on each block and check that your answers make sense.

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

Pick one card you put under 'We DON'T know this yet.' What would you have to figure out first to move it to the known column, and which quantity does it depend on?

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